

Лабораторная работа №2

Структуры данных

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```
PS C:\Windows\system32> ssh -Y mininet@192.168.56.3
mininet@192.168.56.3's password:
Welcome to Ubuntu 20.04.1 LTS (GNU/Linux 5.4.0-42-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings

Last login: Wed Sep 23 06:27:04 2026 from 192.168.56.1
mininet@mininet-vm:~$
```

Обновлены репозитории программного обеспечения.

```
mininet@mininet-vm:~$ sudo apt-get update
Err:1 http://security.ubuntu.com/ubuntu focal-security InRelease
      Temporary failure resolving 'security.ubuntu.com'
Err:2 http://us.archive.ubuntu.com/ubuntu focal InRelease
      Temporary failure resolving 'us.archive.ubuntu.com'
Err:3 http://us.archive.ubuntu.com/ubuntu focal-updates InRelease
      Temporary failure resolving 'us.archive.ubuntu.com'
Err:4 http://us.archive.ubuntu.com/ubuntu focal-backports InRelease
      Temporary failure resolving 'us.archive.ubuntu.com'
Reading package lists... Done
```

```
mininet@mininet-vm:~$ iperf3 --version
iperf 3.7 (cJSON 1.5.2)
Linux mininet-vm 5.4.0-42-generic #46-Ubuntu SMP Fri Jul 10 00:24:02 UTC 2020 x86_64
Optional features available: CPU affinity setting, IPv6 flow label, SCTP, TCP congestion algorithm setting, sendfile / zero-copy, socket pacing, authentication
mininet@mininet-vm:~$
```

Установлено вспомогательное ПО: git, jq, gnuplot-nox, evince.

```
mininet@mininet-vm:~$ sudo apt-get install -y git jq gnuplot-nox evince
Reading package lists... Done
Building dependency tree
Reading state information... Done
gnuplot-nox is already the newest version (5.2.8+dfsg1-2).
evince is already the newest version (3.36.10-0ubuntu1).
git is already the newest version (1:2.25.1-1ubuntu3.14).
jq is already the newest version (1.6-1ubuntu0.20.04.1).
```

```
mininet@mininet-vm:~$ which plot_iperf.sh
/usr/bin/plot_iperf.sh
mininet@mininet-vm:~$ _
```

Задана простейшая топология из двух хостов и коммутатора.

```
mininet@mininet-vm:~$ sudo mn --topo=single,2 -x
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1)
*** Configuring hosts
h1 h2
Error starting terms: Cannot connect to display
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Starting CLI:
mininet>
```

```
mininet> net
h1 h1-eth0:s1-eth1
h2 h2-eth0:s1-eth2
s1 lo: s1-eth1:h1-eth0 s1-eth2:h2-eth0
c0
mininet> links
h1-eth0<->s1-eth1 (OK OK)
h2-eth0<->s1-eth2 (OK OK)
mininet> dump
<Host h1: h1-eth0:10.0.0.1 pid=753>
<Host h2: h2-eth0:10.0.0.2 pid=756>
<OVSSwitch s1: lo:127.0.0.1,s1-eth1:None,s1-eth2:None pid=761>
<Controller c0: 127.0.0.1:6653 pid=746>
mininet>
```

Запуск сервера, затем клиента

```
mininet> h2 iperf3 -s &
mininet> h1 iperf3 -c 10.0.0.2 -u
Connecting to host 10.0.0.2, port 5201
[ 5] local 10.0.0.1 port 49115 connected to 10.0.0.2 port 5201
[ ID] Interval            Transfer      Bitrate      Total Datagrams
[ 5] 0.00-1.00 sec       129 KBytes   1.05 Mbits/sec  91
[ 5] 1.00-2.00 sec       127 KBytes   1.04 Mbits/sec  90
[ 5] 2.00-3.00 sec       129 KBytes   1.05 Mbits/sec  91
[ 5] 3.00-4.00 sec       127 KBytes   1.04 Mbits/sec  90
[ 5] 4.00-5.00 sec       129 KBytes   1.05 Mbits/sec  91
[ 5] 5.00-6.00 sec       129 KBytes   1.05 Mbits/sec  91
[ 5] 6.00-7.00 sec       127 KBytes   1.04 Mbits/sec  90
[ 5] 7.00-8.00 sec       129 KBytes   1.05 Mbits/sec  91
[ 5] 8.00-9.00 sec       127 KBytes   1.04 Mbits/sec  90
[ 5] 9.00-10.00 sec      129 KBytes   1.05 Mbits/sec  91
-----
[ ID] Interval            Transfer      Bitrate      Jitter      Lost/Total Datagrams
[ 5] 0.00-10.00 sec      1.25 MBytes   1.05 Mbits/sec  0.000 ms    0/906 (0%) sender
[ 5] 0.00-10.00 sec      1.25 MBytes   1.05 Mbits/sec  0.011 ms    0/906 (0%) receiver

iperf Done.
mininet> 
```

Тест с изменением порта.

```
mininet> h1 iperf3 -c 10.0.0.2 -p 3250
Connecting to host 10.0.0.2, port 3250
[ 5] local 10.0.0.1 port 56102 connected to 10.0.0.2 port 3250
[ ID] Interval          Transfer    Bitrate    Retr  Cwnd
[ 5]  0.00-1.00    sec   3.02 GBytes  25.9 Gbits/sec    0   8.10 MBytes
[ 5]  1.00-2.00    sec   3.33 GBytes  28.5 Gbits/sec    0   8.10 MBytes
[ 5]  2.00-3.00    sec   3.38 GBytes  29.0 Gbits/sec    0   8.10 MBytes
[ 5]  3.00-4.00    sec   3.27 GBytes  28.1 Gbits/sec    0   8.10 MBytes
[ 5]  4.00-5.00    sec   3.58 GBytes  30.8 Gbits/sec    0   8.10 MBytes
[ 5]  5.00-6.00    sec   3.28 GBytes  28.2 Gbits/sec    0   8.10 MBytes
[ 5]  6.00-7.00    sec   2.88 GBytes  24.7 Gbits/sec    0   8.10 MBytes
[ 5]  7.00-8.00    sec   3.00 GBytes  25.8 Gbits/sec    0   8.10 MBytes
[ 5]  8.00-9.00    sec   3.43 GBytes  29.4 Gbits/sec    0   8.10 MBytes
[ 5]  9.00-10.00   sec   3.35 GBytes  28.8 Gbits/sec    0   8.10 MBytes
-----
[ ID] Interval          Transfer    Bitrate    Retr
[ 5]  0.00-10.00   sec   32.5 GBytes  27.9 Gbits/sec    0
[ 5]  0.00-10.00   sec   32.5 GBytes  27.9 Gbits/sec    0
                                     sender
                                     receiver

iperf Done.
mininet>
```

```
mininet@mininet-vm:~/work/lab_iperf3$ cd ~/work/lab_iperf3
mininet@mininet-vm:~/work/lab_iperf3$ ls -l
total 16
-rw-r--r-- 1 mininet mininet 7795 Sep 24 04:10 iperf3_results.json
-rw-rw-r-- 1 mininet mininet  951 Sep 24 04:13 iperf.csv
drwxrwxr-x 2 mininet mininet 4096 Sep 24 04:13 results
mininet@mininet-vm:~/work/lab_iperf3$ ls -l results/
total 4
-rw-rw-r-- 1 mininet mininet 483 Sep 24 04:13 1.dat
mininet@mininet-vm:~/work/lab_iperf3$
```

Завершение работы

```
mininet@mininet-vm:~/work/lab_iperf3$ sudo mn -c
*** Removing excess controllers/ofprotocols/ofdatapaths/pings/noxes
killall controller ofprotocol ofdatapath ping nox_core lt-nox_core ovs-openflowd ovs-controller ovs-testcontroller
test mnexec ivs ryu-manager 2> /dev/null
killall -9 controller ofprotocol ofdatapath ping nox_core lt-nox_core ovs-openflowd ovs-controller ovs-testcontroller
pbwtest mnexec ivs ryu-manager 2> /dev/null
pkill -9 -f "sudo mnexec"
*** Removing junk from /tmp
rm -f /tmp/vconn* /tmp/vlogs* /tmp/*.out /tmp/*.log
*** Removing old X11 tunnels
*** Removing excess kernel datapaths
ps ax | egrep -o 'dp[0-9]+' | sed 's/dp/nl:/'
*** Removing OVS datapaths
ovs-vsctl --timeout=1 list-br
ovs-vsctl --timeout=1 list-br
*** Removing all links of the pattern foo-ethX
ip link show | egrep -o '([_-.[:alnum:]]+-eth[[:digit:]]+)'
ip link show
*** Killing stale mininet node processes
pkill -9 -f mininet:
*** Shutting down stale tunnels
pkill -9 -f Tunnel=Ethernet
pkill -9 -f .ssh/mn
rm -f ~/.ssh/mn/*
*** Cleanup complete.
mininet@mininet-vm:~/work/lab_iperf3$ exit
logout
Connection to 192.168.56.3 closed.
PS C:\Windows\system32>
```

Итоги работы

1. Изучен инструмент iPerf3 для измерения пропускной способности сети в режиме реального времени.
2. Освоены режимы работы iPerf3: TCP, UDP, с указанием порта, с обработкой одного клиента, с экспортом в JSON.
3. Проведены интерактивные эксперименты в среде Mininet с топологией «коммутатор + два хоста».
4. Средняя пропускная способность TCP-соединения в моделируемой сети составила **23.8 Gbits/sec**, что соответствует высокоскоростному соединению без потерь.
5. Освоена визуализация результатов iPerf3 с помощью iperf3_plotter: получены графики окон перегрузки, повторных передач, RTT, пропускной способности, MTU и объёма переданных данных.
6. В ходе работы выявлена несовместимость актуальной версии iperf3_plotter (требуется Python 3.10+) с Python 3.8 в VM Mininet; задача решена использованием legacy-версии скрипта.